

E-Commerce Customer Analytics

Python Data Analysis Portfolio Project Brief

1. Business Scenario

You are a Data Analyst for a fictional Pakistan-based e-commerce retailer selling across six major product categories through a website, mobile app and marketplaces. Management wants an evidence-based review of 2024-2025 performance before making decisions about retention, marketing spend, product mix and promotions.

2. Your Assignment

Use Python in a Jupyter Notebook to perform the complete analysis. Treat the supplied files as raw operational data. Do not assume they are clean or internally consistent. Document reasoning, cleaning decisions, assumptions and findings in Markdown alongside your code.

3. Dataset Package

File	Rows	Purpose
ecommerce_customers.csv	6,518	Customer profile and acquisition data
ecommerce_products.csv	240	Product hierarchy, price and cost
ecommerce_orders.csv	24,022	Order-level transaction attributes
ecommerce_order_items.csv	50,027	Product-level order lines
ecommerce_returns.csv	788	Return events and reasons
ecommerce_data_dictionary.csv	31	Column definitions

4. Core Business Questions

1. How did revenue, order volume, average order value and gross profit change over time?
2. Which categories and subcategories generate the most revenue and profit, and which underperform?
3. Which cities, sales channels and customer segments contribute most to performance?
4. How concentrated is revenue among customers? Identify high-value and repeat customers.
5. How do acquisition channels differ in customer value, repeat purchasing and order behavior?
6. What effect do discounts and promo codes appear to have on revenue, quantity and gross margin?
7. Where are cancellation and return rates highest, and what patterns or return reasons deserve attention?
8. What seasonal or monthly patterns are visible across the two-year period?
9. Which products or customer groups should management prioritize, investigate or reconsider?
10. What 5-8 actionable recommendations can be supported directly by the analysis?

5. Required Python Skills to Demonstrate

- Pandas: loading, inspection, filtering, sorting, missing values, duplicates, type conversion and string cleaning.
- Pandas: merge/join, groupby/agg, pivot_table, transform and vectorized operations.

- NumPy: conditional logic and numerical/array-based transformations where appropriate.
- Datetime analysis and useful time features.
- Derived metrics such as net revenue, cost, gross profit, gross margin, order value and customer metrics.
- EDA: descriptive statistics, distributions, unusual values and relationships.
- Visualization with Matplotlib and optionally Seaborn.
- Analytical reasoning: distinguish data-quality problems from genuine business outliers.

6. Data Quality Challenge

The raw files intentionally contain a modest number of realistic quality issues. Discover them systematically rather than being told exactly where they are. Keep an audit trail of each issue, its likely impact, and your treatment.

7. Advanced Analysis Requirement

Include at least one analysis beyond standard descriptive EDA. Recommended: RFM-style customer segmentation (Recency, Frequency, Monetary value). Cohort/retention analysis, a statistical comparison, anomaly investigation or another defensible method is also acceptable.

8. Expected Notebook Structure

- Project introduction and objectives
- Imports and data loading
- Initial inspection and data-quality assessment
- Cleaning and preparation
- Feature engineering and table integration
- Exploratory data analysis
- Business-question analysis
- Advanced customer analysis
- Key insights
- Recommendations and limitations

9. Deliverables

Primary deliverable: one polished .ipynb notebook suitable for GitHub/public portfolio viewing. Include the raw CSV files and a README.md. A cleaned CSV may be exported if useful, but the notebook should show how raw data became analysis-ready.

10. Analyst Constraints

- Do not modify raw CSV files manually before loading them into Python.
- Do not remove an outlier only because it looks extreme; investigate and justify.
- Do not use Power BI or Excel for calculations that belong in this Python project.
- Avoid hard-coding final analytical results in Markdown.

- Use functions where repetition makes them useful; avoid unnecessary abstraction.

Portfolio objective: demonstrate an analyst's workflow, not merely Python syntax.